FIT2179 Data Visualisation

Course Notes

COLOUR

14 August 2026

COLOUR SPACES

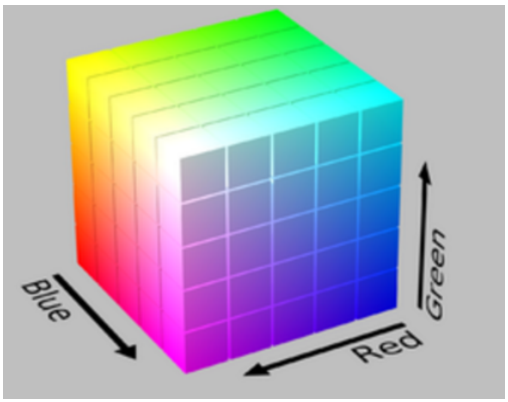
We have seen three colour channels:

- Luminance varies between dark black and bright white. Luminance is also called value or brightness.
- Saturation is how far away a colour is from grey. A desaturated colour is grey and a fully saturated colour is a pure colour.
- Hue is the position in the spectrum of a rainbow.

For storing, displaying, and computing we need colour spaces. Colour spaces are not necessarily based on concepts like hue, saturation and brightness.

RGB

The most common colour space is the RGB colour space. It has three axes and can be represented in a cube. It has a red, a green, and a blue axis. The values along each axis vary between a minimum of 0 and a maximum value, which is identical for all three axis. If we store each value in an 8-bit value, then red green and blue vary between 0 and 255.



0 means absence of light and 255 means a maximum of light. A pure red colour has an R value of 255 and a G and a B value of 0.

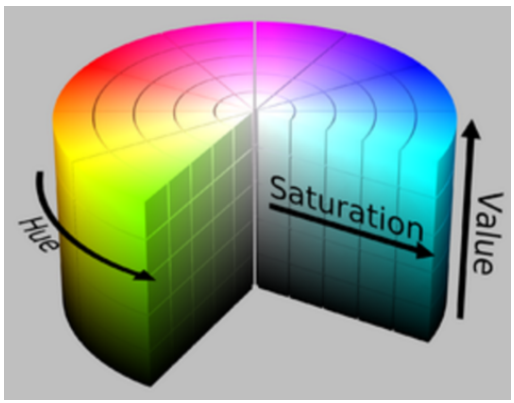
If all values equal 0, the result is black. Black is placed at the corner of the cube above that we cannot see. If all values are 255, the result is white. This is the corner pointing at us.

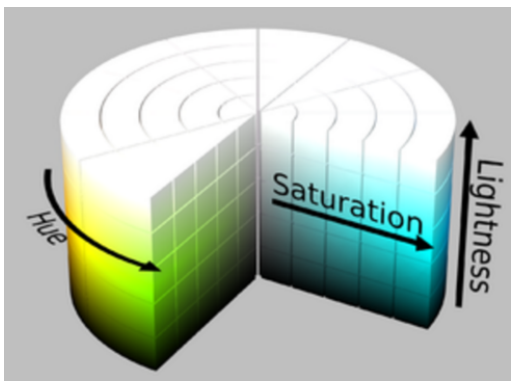
The RGB colour space is used by computer monitors, web browsers, scanners, and digital cameras. However, it is not a convenient colour space to pick colours in a user interface, for example, when designing a data visualisation.

HSV, HSL, HSB

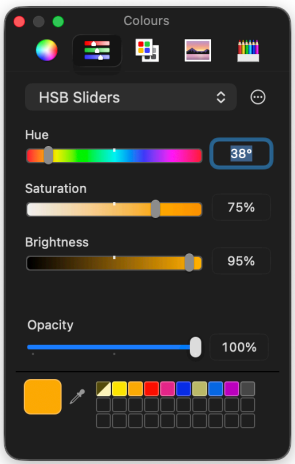
The HSV, HSL and HSB colour spaces are made for picking colours in a user interface. They transform the RGB colour space by mathematical transformation to three intuitive colour channels. HSV, HSL and HSB can be presented as cylinders or cones, and they differ by how they transform from and to

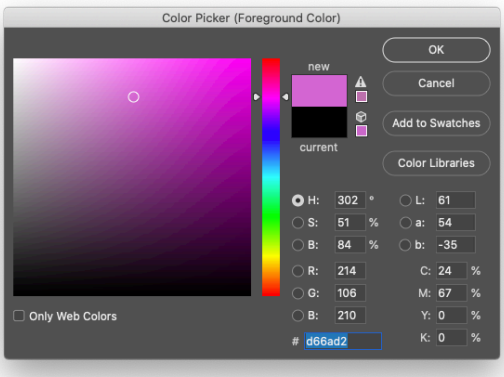
the RGB colour space. H and S stand for hue and saturation, and V, L and B for value, lightness, and brightness.





Below are two colour pickers using the HSB colour space. The first one uses three sliders for hue, saturation and brightness. The second picker is from Adobe Photoshop. The two-dimensional panel varies saturation along the horizontal axis and brightness along the vertical axis. Hue is selected along the vertical rainbow scale. The two channels displayed in the two-dimensional panel can be selected.





PERCEPTUALLY LINEAR OR UNIFORM COLOUR SPACES

The RGB colour space and the derived HSV, HSL or HSB colour spaces are not perceptually linear or uniform colour spaces.

What does that mean? Assume that you move by a distance of 50 units in the RGB colour space. Depending on where you start inside the RGB cube, the colour will change massively, or it will only change a little bit.

This is a problem when we need to compute with colour values, for example, when interpolating between two colours. When we compute the numerical mean value of two colours, we also expect this colour to be visually between the two original colours.

Colour management also requires uniform colour spaces. Colour management transforms colour values between devices with different colour properties, for example, from a video camera to a monitor, or from a scanner to a printer.

A perceptual linear or uniform colour space is a colour model that seeks to make the colour-making attributes perceptually uniform, i.e. identical spatial distance between two colours equals identical amount of perceived colour difference.

Source: Wikipedia Color Appearance Model

CIELAB OR L*a*b*

The L*a*b* colour space is a perceptually linear colour space. If we compute the component-wise mean value of two colours in this colour space, the resulting mean value is also visually between the two original colours. There are three axes:

- L*: luminance axis
- a: green–red axis
- b: yellow–blue axis

The a axis varies between green and red. The b axis varies between yellow and blue.

Note that values can be negative. Green is negative and yellow is negative. If the a value and the b value are zero, then we get a colourless, or grey colour.

DESIGNING DATA VISUALISATIONS WITH COLOUR

How should we choose colour when designing data visualizations? To answer this question, read the excellent blog post What to consider when choosing colors for data visualization by Lisa Charlotte Muth at Datawrapper. The following ideas and figures are copied from her blog post.

WHEN TO USE COLOUR

The first recommendation is to not overuse colour. For example, we should not use more than a handful of hues to indicate different categories. If we use too many different colour hues, it becomes very difficult to remember and match colours with attribute values. In fact, in the example below it is better to not use colour at all, and replace the legend with labels and annotations.



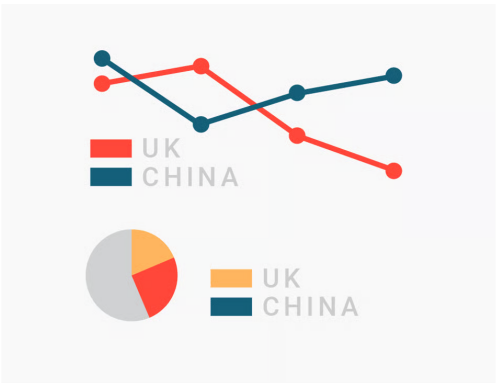
NOT IDEAL



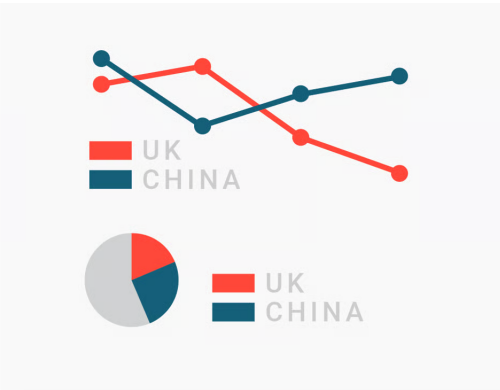
BETTER

SAME COLOUR SAME VARIABLE

The second advise is to use the same colours for the same variables. In the first visualization below, the blue colour is used inconsistently. In the improved visualisation, blue is used consistently for information about China.



NOT IDEAL



BETTER

COLOUR CONTRAST

Make sure the contrast between foreground and background colour is visually strong enough. For example, gray text on a blue background is difficult to read.



CONTRAST RATIOS

INTUITIVE COLOURS

Use colours that your readers intuitively understand. Green means *yes*, or *good*. Red means *no*, *stop* or *bad*. Also prefer colour that can be observed in real life, such as, green for forest, and blue for water. Avoid stereotypes when selecting colour, such as for different human races, religion, or gender. For example, avoid pink for girls and blue for boys.



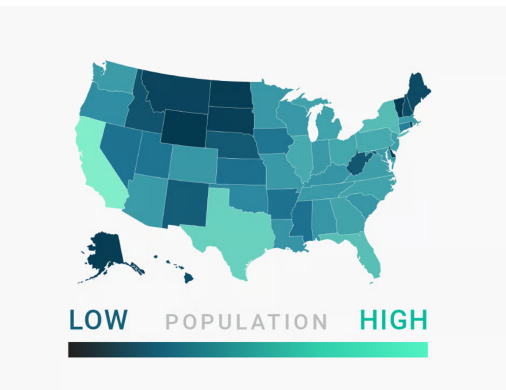
NOT IDEAL



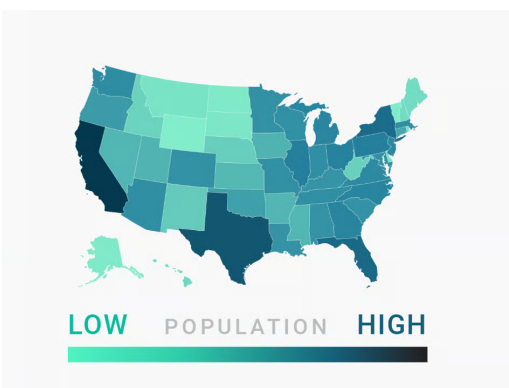
BETTER

LIGHT IS LOW, DARK IS HIGH

Use bright colours for low values and dark colours for high values. This is intuitive for readers, but there are exceptions. One exception is when using a very dark background. With a dark background, bright colours can be used to indicate high values. Another exception applies to elevation on topographic maps, where higher values are commonly shown with brighter colours.



NOT IDEAL



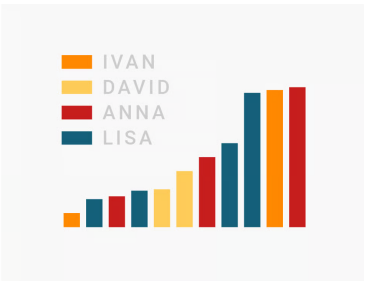
BETTER

HUE IS FOR CATEGORIES

Avoid using saturation and brightness for showing different categorical attribute values. The main channel should be hue.

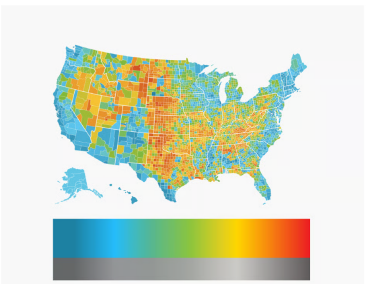


NOT IDEAL

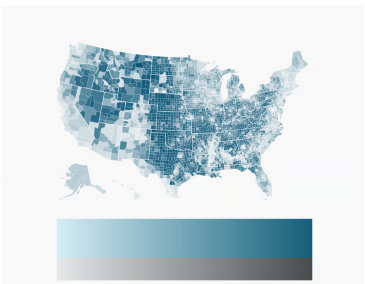


BETTER

The same applies when using a colour gradient for a quantitative attribute. Use lightness as the primary channel and optionally vary between two hues. See the page on the *evil rainbow colour palette* later in this script.



NOT IDEAL



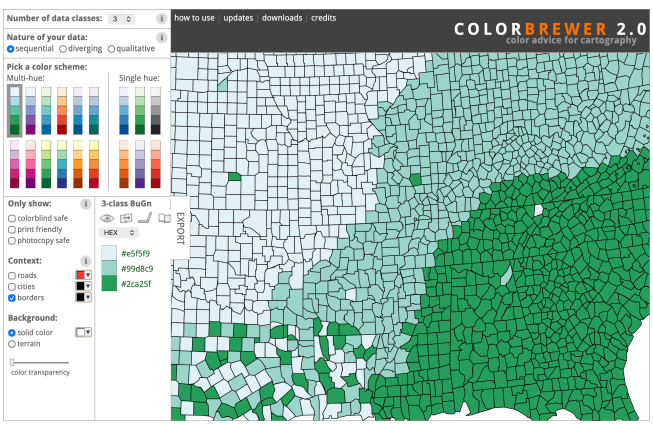
BETTER

COLORBREWER

Building good colour gradients is difficult; use ColorBrewer instead. ColorBrewer lets you select the number of colours, the colour hue, and whether you need a sequential, diverging or qualitative colour scheme.

ColorBrewer can also suggest colours that are visible to people with colour vision impairment.

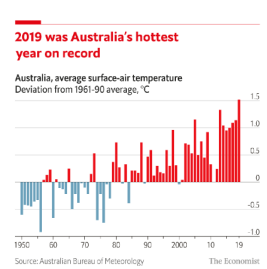
https://colorbrewer2.org/



COLOUR CONSISTENCY

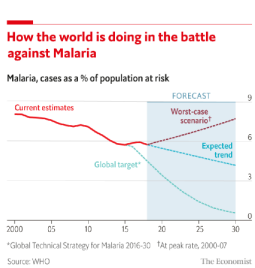
Consistency in colour use also helps link multiple visualizations and build a brand. Below are visualisations created for the Economist website. You can see that all visualisations consistently use the same red colour and a pale blue colour.

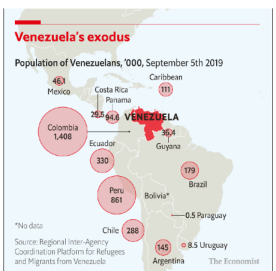
Also note that all visualization only use a small number of different colours. This results in elegant, recognisable, and easy-to-read visualisations.



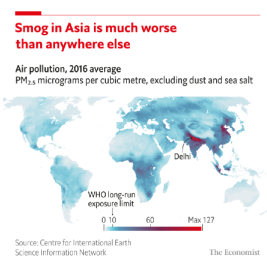


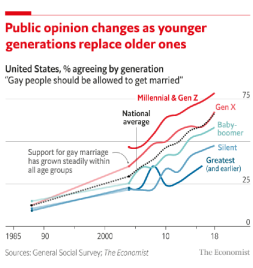


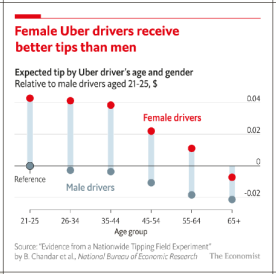


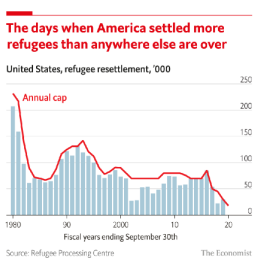




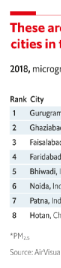


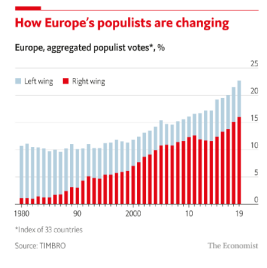




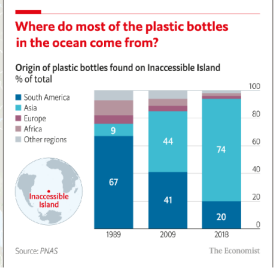












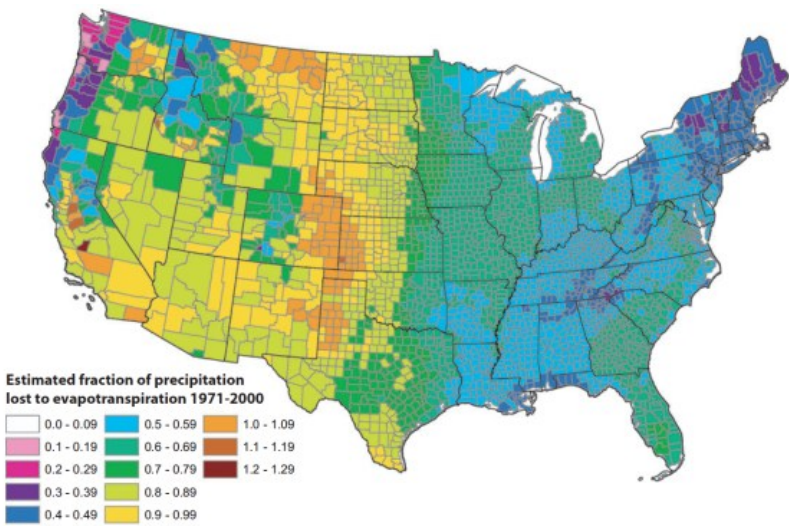




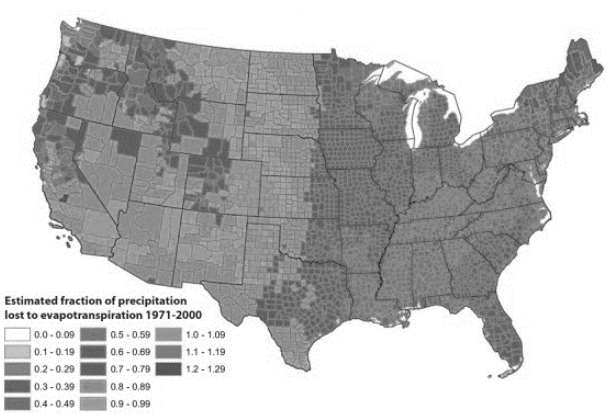


THE EVIL RAINBOW COLOUR PALETTE

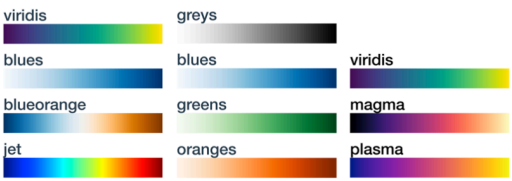
The map below uses one of many different rainbow colour palettes that mainly vary hue for showing quantitative information.



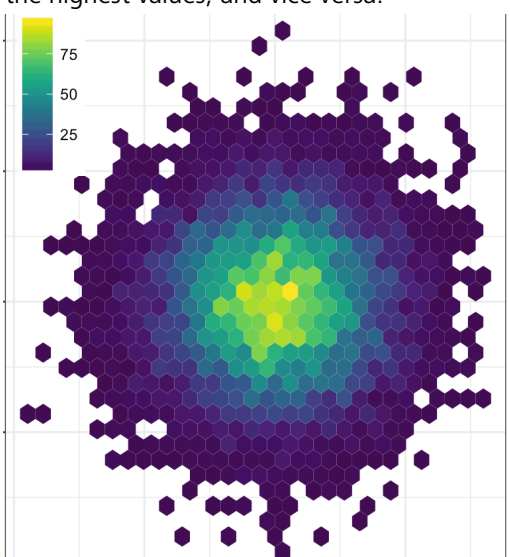
This palette is an extremely poor choice, because it is difficult to interpret. When converted to greyscale, it is obvious that brightness is not applied consistently. Light colours do not consistently indicate low values and vice versa.



Various user studies have been conducted to find and evaluate colourful and effective alternatives to rainbow colour palettes.



A study by Liu and Heer (2018) found that palettes with a single hue work well. Varying hue within limits also works well when combined with changing luminance. A successful example is the Viridis palette. It varies hue among blue, green and yellow, but the main visual channel is still brightness. The brightest colours in Viridis show the highest values, and vice versa.



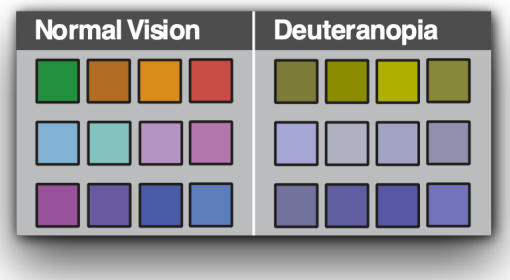
Liu, Y. and Heer, J. (2018).
Somewhere over the rainbow: An empirical assessment of quantitative colormaps. 2018 CHI Conference on Human Factors in Computing Systems (p. 598). ACM.

COLOUR VISION DEFICIENCY

Colour vision deficiency or colour blindness affects about 8% of all men. Women are rarely affected, because the condition is linked to the X chromosome.

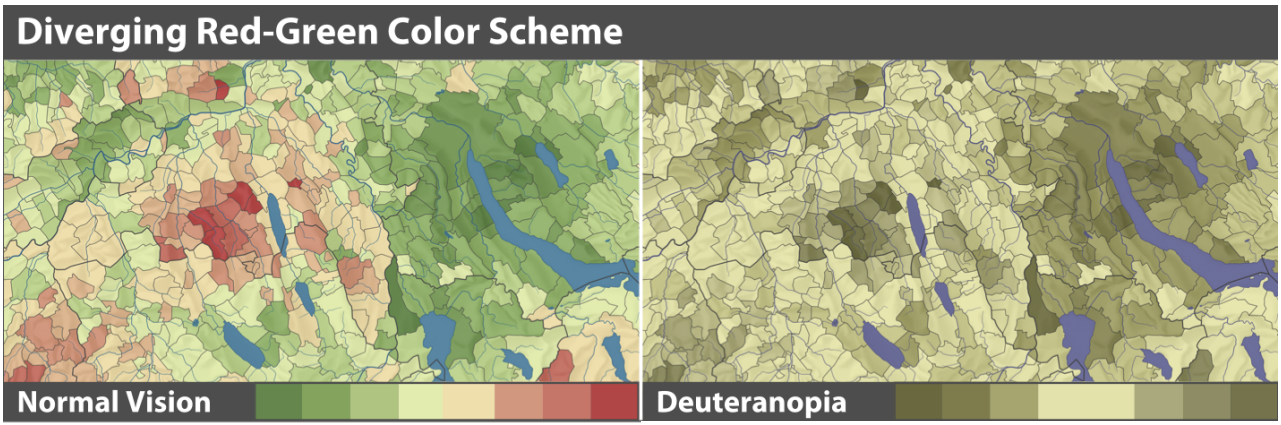
A variety of colour vision deficiencies exist, but most affected people have difficulty distinguishing red and green. The term colour-blind is misleading, because they perfectly see other colours, and most can still distinguish shades of red and green to a certain degree.

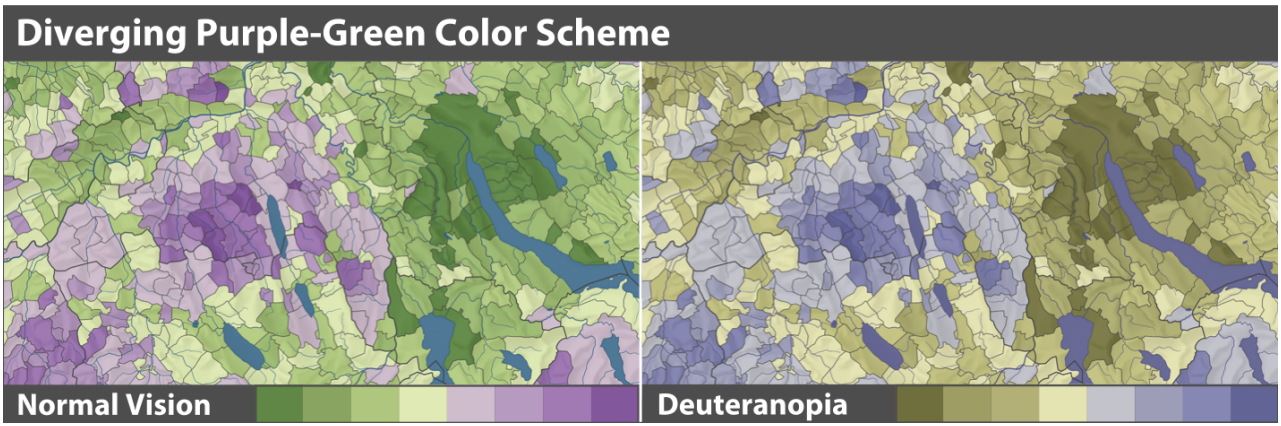
The figure below shows how someone with colour vision impairment perceives colour. On the left are a few colours that someone with normal vision can distinguish. On the right are the same colours as seen by someone with deuteranopia, a strong red-green vision impairment. If the two sides look very similar or even identical to you, then you have the deuteronopia condition.



Red-green palettes are particularly problematic in data visiualisation. On the next page is an example of a map showing no votes in red and yes votes in green. On the right side is the map as someone with colour blindness sees it. The yes and no votes are indistinguishable.

Below is the same map with a different colour palette. The red colours are replaced with purple colours, which are seen as blue colours by those with colour vision impairment.





COLOR ORACLE

Software can simulate colour blind vision, which is useful to make sure data visualizations are readable by everybody. An example is the Color Oracle app.

colororacle.org